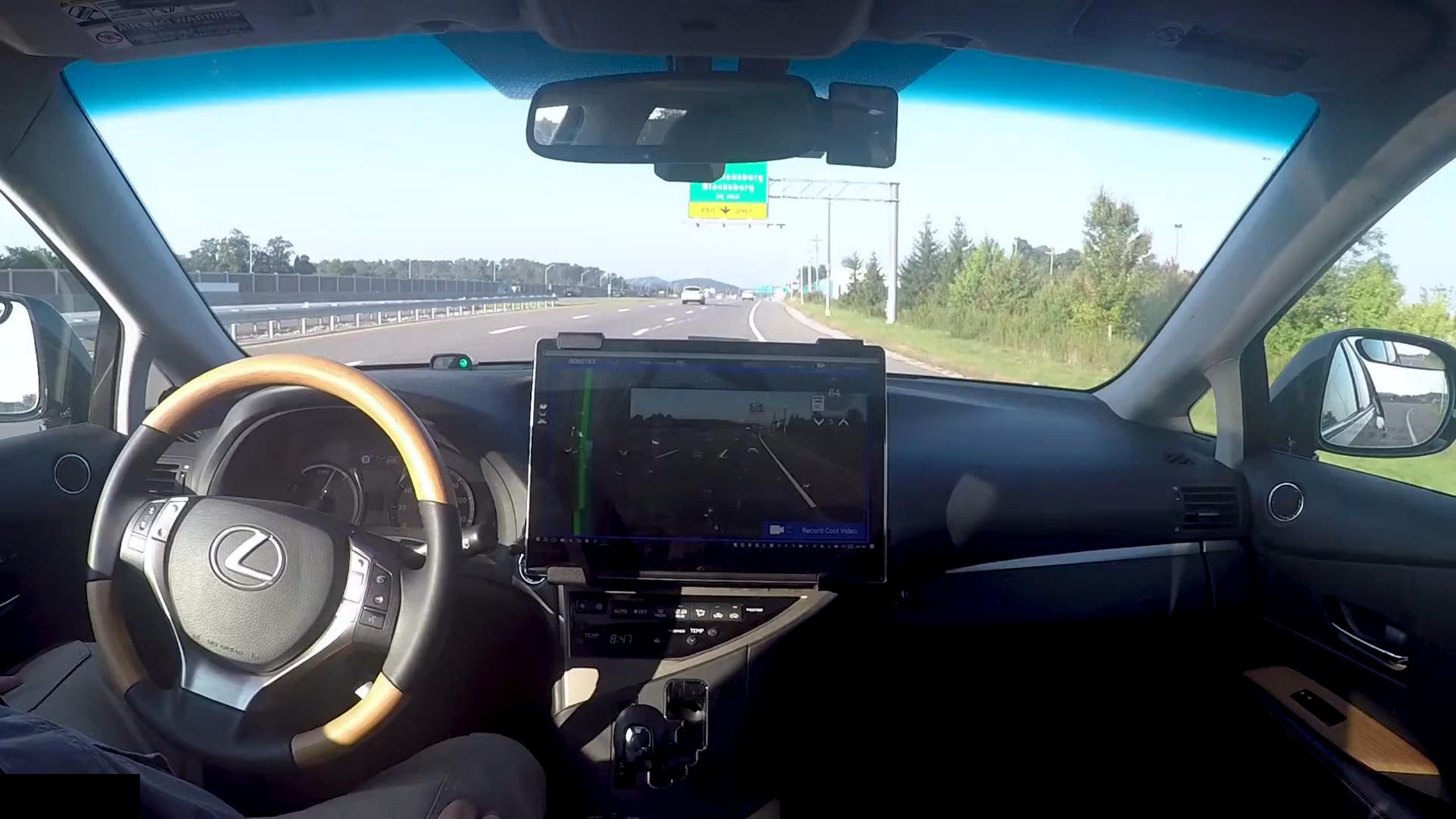


Safety

Dylan Losey



ROBOT AREA SAFETY



This Lecture



- What does it mean for a robot to be safe?
- How do we balance between safety and performance?

Defining Safety

I think of safety as a **constraint**. Something that we must enforce, regardless of the robot's other objectives.

- Examples of safety include:
 - Collisions between the human and the robot
 - Critical task failures (e.g., a drone colliding with a wall)
 - Altering the environment in an undesired way (e.g., spilling a liquid across the table)

Defining Safety

I think of safety as a **constraint**. Something that we must enforce, regardless of the robot's other objectives.

At a high-level, we can formulate safety along the lines:

$$a \leftarrow \operatorname{argmax} (\text{complete my task})$$

$$\text{s. t. } a \in (\text{safe set})$$

Defining Safety

I think of safety as a **constraint**. Something that we must enforce, regardless of the robot's other objectives.

When determining if an action is **safe**, we need to consider both:

- its immediate outcomes (e.g., hitting a wall) and
- its long-term outcomes (e.g., speeding up so the robot cannot avoid a wall)

How would you design an algorithm for safe human-robot interaction?



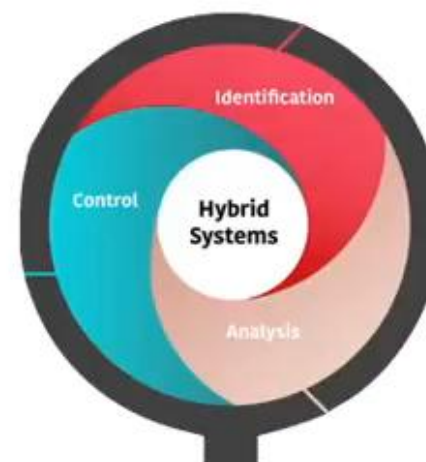
One approach to
safety is to reason
over the likely
human actions.





Probabilistically Safe Robot Planning with Confidence-Based Human Predictions

Jaime F. Fisac,* Andrea Bajcsy,* Sylvia L. Herbert, David Fridovich-Keil,
Steven Wang, Claire J. Tomlin, and Anca D. Dragan



Safety Filters

Given task τ , safety metric f , and human model P

For each timestep:

- Use the human model to predict the human's future actions ξ_H (this could be deterministic or stochastic trajectory)
- Perform constrained optimization:
Find the sequence of robot actions that maximize task performance while (probabilistically) keeping the safety metric above some threshold

$$\xi_R \leftarrow \operatorname{argmax} R(\xi, \xi_H)$$

$$\text{s. t. } f(\xi, \xi_H) \geq 0$$

- Take the first action(s) in this trajectory and repeat

Human Model

$$P(a \mid s, \theta) \propto e^{\beta \cdot Q_{\theta}(s,a)}$$

β captures how **rational** the human is.

- When $\beta \rightarrow \infty$, the human always takes the optimal action (*perfectly rational*)
- When $\beta \rightarrow 0$, the human takes random actions (*uniform distribution*)

Human Model

$$P(a \mid s, \theta) \propto e^{\beta \cdot Q_{\theta}(s,a)}$$

Tells us how the human will act **now**...

...but we want to know how the human will act **over time**

Human Model

Initialize D as empty set

For each iteration:

- Initialize s_H at human's current state

- For T timesteps:

- Sample action $a_H \sim P(a \mid s_H, \theta)$

- Update human state $s_H^{t+1} = f(s_H^t, a_H)$

- Add to dataset $D = D + s_H^{t+1}$

Human Model

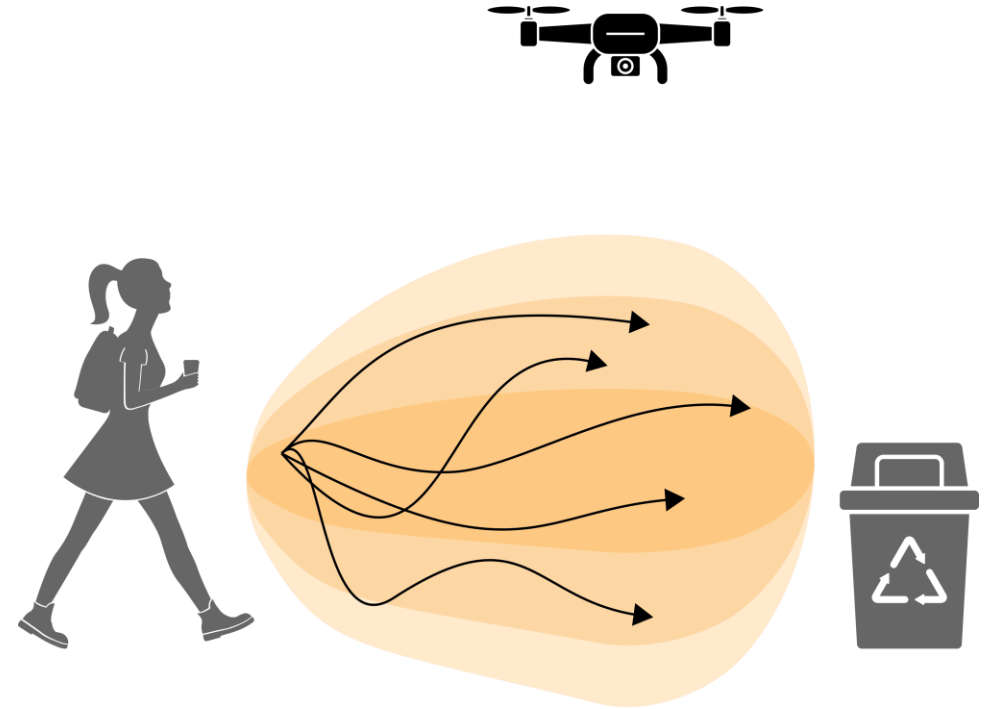
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- Add to dataset $D = D + s_H^{t+1}$



Probabilistically Safe

We do not know exactly where the human will be. Hence, we cannot guarantee there will *never* be a collision.

$$\xi_R = \operatorname{argmax}_{\xi \in \Xi} R(\xi)$$

Trajectory optimization over robot's objective

$$P_{\text{collision}}(\xi, D) \leq \sigma$$

Subject to the constraint that the probability of collision with human is $\leq \sigma$

Probabilistically Safe

$$P_{collision}(\xi, D) \leq \sigma$$

$$P_{collision}(\xi, D) \leftarrow 0$$

For every timestep t in T :

$s_R^t \leftarrow \xi(t)$ get robot position at timestep

For every $s_H^t \in D$:

- Check for collision

Probabilistically Safe

$$P_{collision}(\xi, D) \leq \sigma$$

$$P_{collision}(\xi, D) \leftarrow 0$$

For every timestep t in T :

$s_R^t \leftarrow \xi(t)$ get robot position at timestep

For every $s_H^t \in D$:

- Check for collision

$$P_{collision}(s_R^t, D) \leftarrow \text{number of collisions} / \text{number of } s_H^t \text{ states}$$

$$P_{collision}(\xi, D) \leftarrow \max\left(P_{collision}(\xi, D), P_{collision}(s_R^t, D)\right)$$

Probabilistically Safe

The resulting robot trajectory will vary based on:
The human's **rationality** β and the **threshold** σ

$$\xi_R = \operatorname{argmax}_{\xi \in \Xi} R(\xi)$$

Trajectory optimization over robot's objective

$$P_{\text{collision}}(\xi, D) \leq \sigma$$

Subject to the constraint that the probability of collision with human is $\leq \sigma$

Takeaways

The flavor of the **optimization function** and the **constraint function** will vary based on your application. But the key ideas are:

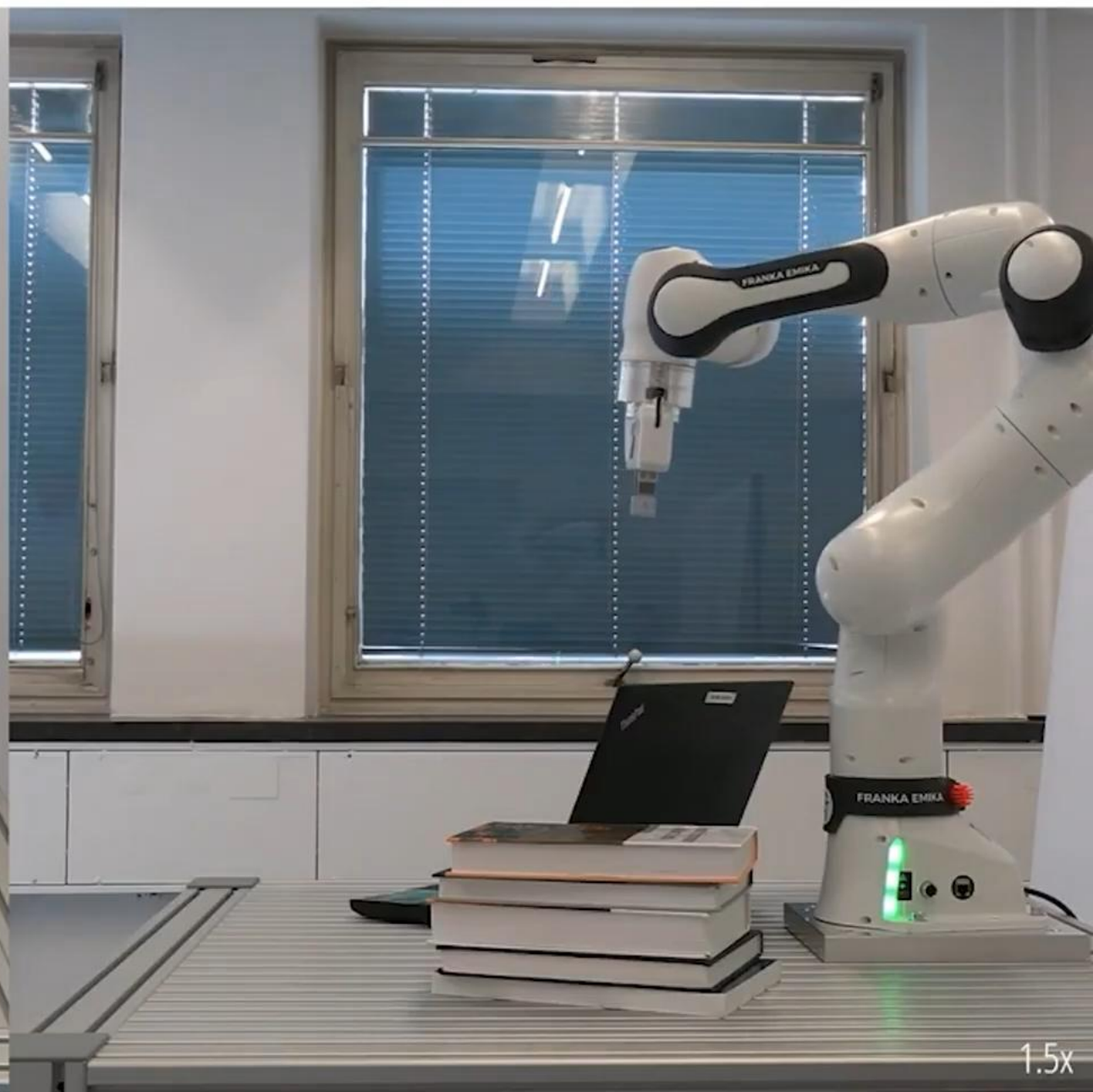
1. You have some way to predict what the human will do.
Could be simple (they will keep moving in same direction) could be complex (game theoretic)
2. You have a metric to measure safe / unsafe
Check whether a given state is safe or unsafe (what does it mean to be safe?)
3. You optimize for performance, while making sure your metric is satisfied
Methods like MPC are fine, just may need to adjust the output actions to satisfy constraint



One fundamental
trade-off revolves
around how
conservative we
want to be.



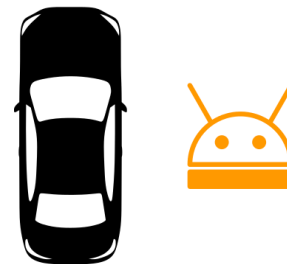
Without holding an object, the robot can safely move over the laptop.



Motivating Problem

Robot car trying to **pass** a human driver.

- Human *does not* think about robot
- Robot does not know exactly where the human will go or what the human will do
- The robot has a **distribution** $M(\xi_H)$ over the possible human trajectories



An aerial photograph of a multi-lane highway with several vehicles. Overlaid on the image are numerous concentric yellow circles of varying sizes, centered on each vehicle. These circles represent the range of sensors (like radar or lidar) used by the vehicles to detect their surroundings. The circles overlap significantly, illustrating the concept of a vehicle's 'risk-aware' perception of its environment. The text 'What is risk-aware control?' is superimposed in the center of the image.

What is **risk-aware**
control?

Risk-Aware Control

$$\xi_R = \operatorname{argmax}_{\xi \in \Xi} R(\xi, \xi_H)$$

We want to find a robot trajectory that **maximizes** the reward.

Given the exact human trajectory $\xi_H = (s_H^0, \dots, s_H^T)$, this is easy...

Risk-Aware Control

$$\xi_R = \operatorname{argmax}_{\xi \in \Xi} R(\xi, \xi_H)$$

We want to find a robot trajectory that **maximizes** the reward.

Given the exact human trajectory $\xi_H = (s_H^0, \dots, s_H^T)$, this is easy...

... but what if, instead there is a **distribution** $\xi_H \sim M(\cdot)$

Risk-Aware Control

$$\xi_R = \operatorname{argmax}_{\xi \in \mathbb{E}} \mathbb{E}_{\xi_H \sim M(\cdot)} R(\xi, \xi_H)$$

*Robot considers how its trajectory will work in expectation
i.e., average performance across the distribution*

Risk-Aware Control

$$\xi_R = \operatorname{argmax}_{\xi \in \Xi} \mathbb{E}_{\xi_H \sim M(\cdot)} R(\xi, \xi_H)$$

$$\xi_R = \operatorname{argmax}_{\xi \in \Xi} \frac{1}{N} \sum_{i=1}^N R(\xi, \xi_{H,i})$$

*Approximation. Sample N trajectories from $M(\cdot)$ and maximize **average reward** across those samples.*

An aerial photograph of a multi-lane highway with several cars. Overlaid on the image are numerous yellow concentric circles of varying radii, centered on each car. These circles represent the range of sensors (like radar or lidar) on the vehicles, illustrating how they perceive their environment. The circles overlap significantly, showing the interconnectedness of the vehicles' fields of view.

What if we focused on the
worst-case instead of the
average?

Risk-Averse Control

$$\xi_R = \operatorname{argmax}_{\xi \in \Xi} \min_{\xi_H \sim M(\cdot)} R(\xi, \xi_H)$$

Inner loop: *find the worst case if we follow ξ*

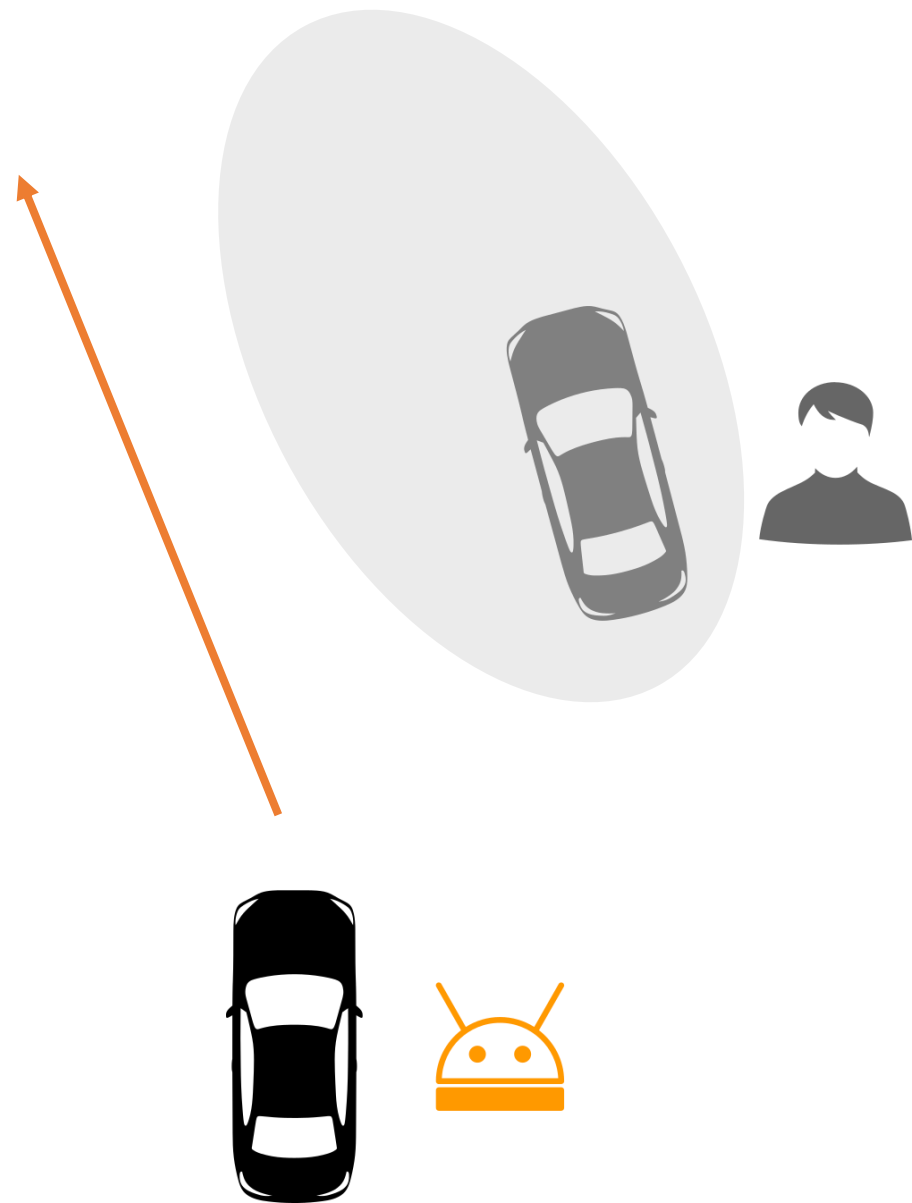
Outer loop: *find the ξ that maximizes the worst case performance*

Risk-Averse Control

$$\xi_R = \operatorname{argmax}_{\xi \in \Xi} \min_{\xi_H \sim M(\cdot)} R(\xi, \xi_H)$$

$$\xi_R = \operatorname{argmax}_{\xi \in \Xi} \left(\min_{\xi_{H,1} \dots \xi_{H,N}} R(\xi, \xi_{H,i}) \right)$$

Approximation. Sample N trajectories from $M(\cdot)$, then alternate between the min inner loop and max outer loop.



An aerial photograph of a multi-lane highway with several cars. Overlaid on the image are numerous yellow concentric circles of varying radii, centered on each car. These circles represent the range of sensors (like radar or lidar) on the vehicles, illustrating how they perceive their environment. The circles overlap significantly, showing the interconnectedness of the vehicles' sensor fields. The text "What if we focused on the best-case?" is superimposed in the center of the image.

What if we focused on the
best-case?

Risk-Seeking Control

$$\xi_R = \operatorname{argmax}_{\xi \in \Xi} \max_{\xi_H \sim M(\cdot)} R(\xi, \xi_H)$$

Inner loop: *find the best case if we follow ξ*

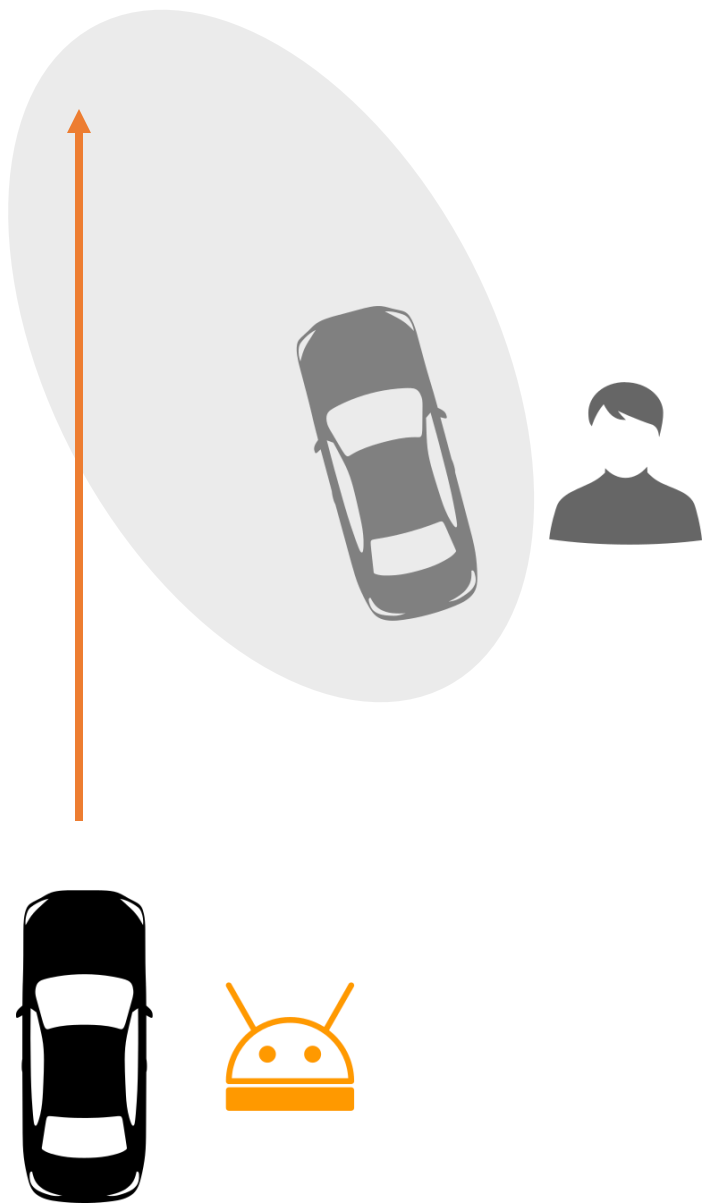
Outer loop: *find the ξ that maximizes the best case performance.*
Only difference from risk-averse is now we **max** in inner loop.

Risk-Seeking Control

$$\xi_R = \operatorname{argmax}_{\xi \in \Xi} \max_{\xi_H \sim M(\cdot)} R(\xi, \xi_H)$$

$$\xi_R = \operatorname{argmax}_{\xi \in \Xi} \left(\max_{\xi_{H,1} \dots \xi_{H,N}} R(\xi, \xi_{H,i}) \right)$$

Approximation. Sample N trajectories from $M(\cdot)$, then alternate between the max inner loop and max outer loop.



Risk-Seeking vs. Risk-Averse

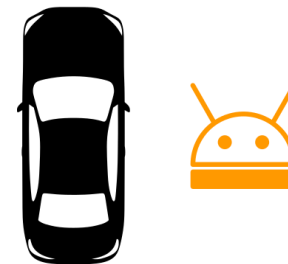
Risk-Averse:

- Robot plays it safe
- Can be too safe (e.g., car never passes)



Risk-Seeking:

- Robot hopes things will work out
- Can be too optimistic (e.g., no yielding to merging human)



Takeaways

Although I want safety to be this constraint we always satisfy, the **stochastic nature of the world makes that impossible**. The only way for our car to be perfectly safe is to never drive (or for the robot arm to never leave its cage).

In human-robot interaction we often end-up trying to **balance** between being robust to humans (i.e., coming up with plans that are safe no matter what) and finding reasonable assumptions we can trust the human to follow (so we reach reasonable plans).

This Lecture



- What does it mean for a robot to be safe?
- How do we balance between safety and performance?

Next Lecture



- How can humans and robots communicate?